



Battle Card, Card Game for Teaching the History of the Incas Through Intelligent Techniques

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Abstract. History plays a vital role in knowing their past and thus improving and not making the same mistakes in the future. However, history teaching has some shortcomings, such as memorizing history without interpretation, which is boring, uncomfortable, and non-interactive when students learn from books and articles. This work aims to cover this gap with the development of Battle Card, a card game based on the Inca empire and implemented with artificial intelligence algorithms, such as the seek and flee and arrive movement algorithms, the algorithm Depth-First Search, and a state machine. The Scrum framework was applied to manage the project and maintain constant communication with the development team. In addition, Phaser.io was used as a framework for game development on desktop and mobile platforms. Tests were carried out, and the results were finally evaluated through game quality metrics. In conclusion, a fun app about the history of the Incas was obtained with an acceptable level of playability that educators and interested parties can use.

Keywords: Card game · Teaching · Incas · Artificial intelligence · History

1 Introduction

History is a fundamental social science that allows us to know our past and improve in the future with the experiences or lessons learned from our ancestors and ourselves [1]. In addition, history allows us to know the place where people live and promote patriotism. Likewise, history education is important, but many times in schools, history is taught so that students have to memorize out of obligation and not because they want to learn.

The education of history has a significant role in our society to develop and create a feeling of patriotism, even more so on the school stage. Children and adolescents find it boring to learn from books, stories, articles. For them, an interactive and effective way of learning is through games with themes related to what they do in class, such as, for example, the history of the Incas, their Quechua language [2], and the conquest of the Spanish, among other topics. In the long term, it is difficult to remember the names of all the Incas unless it is from a game.

To solve the problem of the lack of patriotism and the lack of didactic and interactive teaching in the historical context, it has been necessary to apply the development of a card game.

History is a fundamental social science that allows us to know our past and improve in the future with the experiences or lessons learned from our ancestors and ourselves. In addition, history allows us to know the place where people live. For these reasons, history education has a significant role in developing our society and creating a feeling of patriotism, even more so in the school setting.

However, many times in schools, a traditional model of memorization and repetition is applied. The teacher carries out the lessons, and the students carry out the same activities that are boring for them. In addition, these activities are often carried out 'under pressure' to obtain a grade, rather than being activities of exploration, analysis, reflection, and interpretation [3, 4].

For children and adolescents, an interactive and effective way of learning is through games with themes related to what they do in class, such as, for example, the history of the Incas, their Quechua language, and the conquest of the Spanish, among other topics. In the long term, it is difficult to remember the names of all the Incas unless it is from a game.

To solve the problem of the lack of didactic and interactive teaching in the historical context and to promote patriotism, it has been necessary to apply the development of a card game. The first step in teaching history is to contextualize students about the period's most significant events and characters.

The main objective of this work is to develop a card game [5], "Battle Card," where each card represents the Incas and their detailed information. Having thus, the following specific objectives are (1) Create a game that is capable of dynamically teaching the history of the Empire of the Incas and (2) Identify the influence of a video game as a pedagogical tool in student learning.

Also, this game implements artificial intelligence (A.I.) algorithms so that the game gives the impression that it is being played against a natural person. Still, in reality, it is being played with the same game, such as movement and pathfinding algorithms such as Depth First Search. Likewise, a state machine has been implemented to control the games, cards, and levels' behavior and animation.

For the development of Battle Cards, the Scrum framework was applied for the reason that this project is a short-term project and requires a lot of communication from the team members. In addition, the Trello tool was used to manage the tasks of the different development sub-teams, for example, the design team that is in charge of carrying out the storytelling and the design of the cards, on the other hand, there is the development team which is in order of programming the motion and artificial intelligence algorithms, unit tests, performance, acceptance, and launch of the video game.

The present work is structured as follows: in Section II, the main definitions for the related work are detailed; in Section III, Materials and Methods were made; in Section IV, the results are presented; finally, the conclusions and future works are presented.

2 Related Work

In card games, each player has a deck of cards, where each card symbolizes a character or being that has abilities. The game's objective is to win battles to level up and obtain new cards. Each battle or battle grants specific experience that allows increasing the level; in addition, new cards and more knowledge can be accepted in the case of winning.

According to [6], the game is a fundamental means for the structuring of language and thought; it acts systematically on the psychosomatic balance; enables learning of vital significance; reduces the feeling of gravity in the face of mistakes and failures; invites participation by the player; develops creativity, intellectual competence, emotional strength, and personal stability. In short, it can be affirmed that playing constitutes a fundamental strategy to stimulate the integral development of people in general.

Currently, video games are one of the most used tools in entertainment. Still, not all games are designed to entertain, despite needing specific skills such as adaptability or communication. However, video games focused on education provide the desire to learn, as in [7], where an attractive turn-based card game has been created that aims to incorporate knowledge of mathematics and the history of mythology indirectly. The authors thought that the best way to make a game is based on the currently successful mobile/tablet games. The game is aimed at primary and secondary school students. The proposed educational content is about some concepts in mathematics, such as adding fractions and adding percentages. The game's central theme will focus on mythological history to reveal critical myths. This allows the students to practice mathematics and learn about mythology without feeling that they are doing it while having fun in the game. The implementation technology used is the Unity game engine with the C# programming language to guarantee the deployment of the game on mobiles and tablets. Regarding the results obtained, it is mentioned that it is necessary to make a new face-to-face evaluation to observe the students directly and to be able to verify the effectiveness of the game. According to the user test, the main objective of the video game is to entertain and analyze the gameplay; the authors have achieved the expected result.

Although there are card games related to education as well as the use of engines such as Unity, it must be taken into account that there are games implemented in artificial intelligence, which are widely used when there is only one player against the machine, as in [8], it is a turn-based card game using heuristics implemented in C# language. This is a strategy game where the player and the computer fight in a duel to reduce the opponent's life to 0. Also, the computer is controlled via a GameManager script, which will allow consulting the turn, such as cards that will be played, so to facilitate the best cards, two heuristics have been used, one to maximize the allied side, where the state of the game is valued as attack and health, while the other heuristic is to differentiate between sides and search the maximum advantage for part of the computer. In addition, the Monte Carlo search algorithm was also used, which allows using the best move from among different options.

The paper by Garcia-Sanchez et al. [9] proposes the implementation of an automatic player or autonomous agent capable of playing Hearthstone based on search-based artificial intelligence techniques that, given a game state, are capable of establishing subsequent conditions and taking the best possible action. This work also uses the Monte Carlo and genetic algorithms to computationally adjust good results, improving the agent to

choose the move that leads to a better state without considering states beyond the next. Another work related to this Hearthstone game is [10], where the algorithm's performance is analyzed through parameters such as total wins, time, tree status, heuristics, and the number of iterations.

According to Herreira [11], in his work, he developed a Trading Card Game (TCG) with artificial intelligence; it was created with the Unity game engine; this software has several execution modes: player against player, player vs. A.I., and A.I. vs. A.I. An expert system was developed that emulates the decision-making ability of a human. The Rete algorithm for efficient pattern recognition was applied to implement a rule production system. Good results were obtained in effectiveness and speed. In the same way, [12] he developed the Galactic Arena game that contains an artificial intelligence system capable of playing with any combination of cards. The discrete backpack algorithm was used. This algorithm looks for the best solution among a finite set of possible solutions to a given problem.

On the other hand, artificial intelligence algorithms for finding a solution are important in games. According to [13], in his work, he made a puzzle called Babylon Tower, a three-dimensional puzzle with six discs; each disc consists of six columns of small balls along the side in six different colors. It is mentioned in the article that two methods can be used to find a solution to the Babylon Tower game, namely disk by disk and column by column. The method proposed in this research is the in-depth search column by column.

Also, in the paper by Koracharkornradt [14], he made a block programming game called Tuk Tuk for children of kindergarten, primary and secondary levels. In this game, students can create computer programs in a block-based language to control a car to complete a specific task, earn money, reach the next level, and unlock new coding blocks. Concepts of Depth First Search (DFS) and Breadth-First Search (BFS) have been applied in the game so that children can develop their computational thinking.

3 Materials and Methods

The materials used were case tools that have helped manage the project, mainly we used Trello, and the chosen framework was Scrum, likewise, Artificial Intelligence Algorithms.

3.1 Trello

Trello was used to support the Scrum process [15]. This application bases its methodology on Kanban's Japanese work system, which incorporates boards and cards in a physical workspace to coordinate different activities [16]. Trello helps organize and improve the dynamics of virtual teams since the main attributes it has are its ease of use and versatility.

Each requirement that is defined as a list card must have a name of the requirement, in addition to a label that characterizes that item and can be identified more quickly; the team members who will develop said task must also be attached, is to keep order, priority, and weight of the task.

After performing the tasks, the task is changed to the list; this is to have a new status of the task and see the progress in real-time of all the statuses of the requirements; when the task has been completed, proceed to change to the lists “Done,” which means done or finished.

3.2 Artificial Intelligence Algorithms

Artificial intelligence algorithms will be applied to the proposed game, which will help us solve specific game problems such as the behavior of the cards, that is, player vs. artificial intelligence (A.I). The algorithms to be used are (a) movement algorithms: “Seek and Flee,” “Arrive,” and (b) search algorithms such as “Depth First Search.”

Seek and Flee algorithm. Seek allows you to match a start position to an end position with a specific acceleration. Flee is the opposite of the Seek technique, moving away as fast as possible from the endpoint. The advantage of these algorithms is their complexity, which is $O(1)$.

Arrive algorithm. This algorithm allows an initial position to coincide with a final position, optimally calculating the time so that its arrival is as fast as possible. The advantage of this algorithm is its complexity, which is $O(1)$.

Depth-first search algorithm. This search algorithm allows us to search for an element within a tree or graph. The algorithm starts searching from the root node and branches down only one path until reaching the depths; after that, it returns and continues with the following path until obtaining a search result.

The advantage of DFS is that it requires less memory since only the nodes on the chosen path are stored.

3.3 Scrum

Scrum is a project management framework where people can tackle complex processes and finish on time. In addition, it establishes clear objectives and deadlines, the Scrum methodology is time-based, and each Sprint has a clearly defined objective and duration.

The individual effort of each team member is visible; during the Daily Scrum and Sprint Review meetings, the Product Owner and Scrum Master can track the contribution of each team member during the Sprint.

Scrum improves communication; all team members participate in Scrum meetings and are motivated to express their opinion and contribute to all decisions. The team can easily communicate and remove obstacles that get in the way.

3.4 Sprints

What was done in the sprints is detailed, based on the planning of the project and resolution of the problem.

Sprint 1. The user stories help obtain the product owner’s requirements; they help organize conversationally what the interested party requires; each user story must contain an identifier, the respective name, the priority, the individual description, and the points of acceptance. In our project, we were able to obtain 13 user stories, an example of user stories can be seen in Table 1.

Table 1. User history

Id	HU-001
Name	Start
User	Player
Priority	High
Risk	Medium
Difficulty	5
Description	As a player, I want to enter the new game room to allow me to start playing against the machine
Acceptance	Redirect to the game room when the player presses the start button All associated multimedia elements must be loaded

In the project, the programming in javascript language was developed, and together with the support of the phaser.io framework, avatars were used to represent the opponents (see Fig. 1), assets such as cards or the inverse letter of which you get new cards in the game (see Fig. 2).



Fig. 1. Game main menu.

Artificial Intelligence Techniques. In the project, the game was developed based on artificial intelligence techniques. The first algorithm we produce is ‘seek’ and ‘flee,’ which are in charge of approaching or moving away from the objective. In the following Fig. 3, you can see how the aim is the board, and the card moves to that position [17].

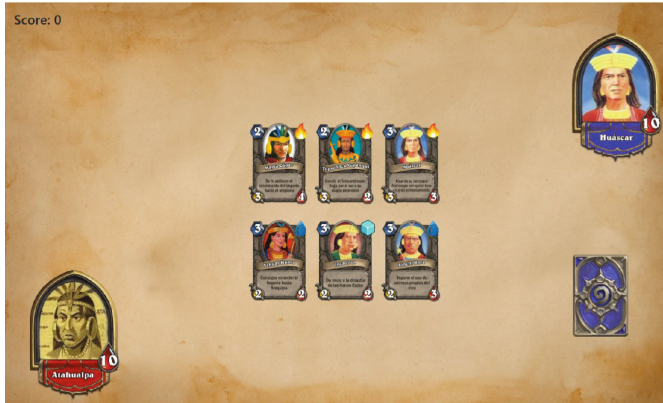


Fig. 2. The initial game of the game.



Fig. 3. Seek move from card to board.

The second algorithm that we develop is the ‘arrive,’ which is in charge of approaching the target. Still, based on a radius, which, if it reaches that radius, increases the speed, in Fig. 4, you can see how the target is the enemy avatar. The card’s effect is attacking the avatar; however, when its impact detects proximity to the target, it increases its speed [17].



Fig. 4. Arrive Movement of the Effect of the Card to the Enemy.

Sprint 2. A redesign was carried out in the cards. What was chosen was to create three cards; the first group is the water type, the second fire, and the last ice; each type has specific characteristics with advantages, unlike the others. The initial design had three powers, and it was decided to add one more power at the top; the cards will have their type represented by an icon in the upper right (see Fig. 5).



Fig. 5. Type cards.

The Battle Rule Between Cards. The game rules were added, where it is described what each card can do and what actions are allowed in the game, some of the rules that may exist when attacking between a card in the deck against an enemy card are shown in Fig. 6.



Fig. 6. Rule of attack of a water card with a fire.

Depth First Search Algorithm. The third algorithm that was used was the Depth-first search; this one is in charge of looking for all the possible routes of each node until its maximum Depth [18]; in the game, it helps to make the best decision looking for the best possible way. These decisions are performed by the machine each shift. To know which is the best route, the algorithm evaluates each one in the rules given by the game; once it has all the results of each branch, it proceeds to assess which is the best option; once this result is obtained, it returns to the primary function the best route found, in Fig. 7, you can see how the algorithm distributes all the cards within a tree. After applying the Depth-first search algorithm, the best route that the algorithm took after analyzing all the results was colored. These decisions are performed by the machine each shift.

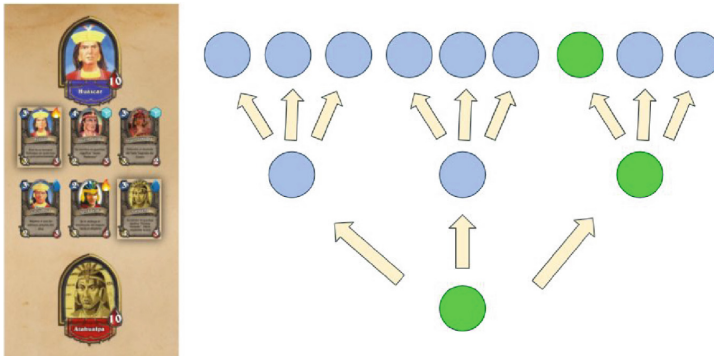


Fig. 7. Execution of the depth first search algorithm.

Sprint 3. A state machine to control the behavior and animations of the video game has been implemented as a decision-making algorithm. In Fig. 8, the states and their transitions are shown according to the player's different actions. Statuses have also been implemented for the game and the levels.

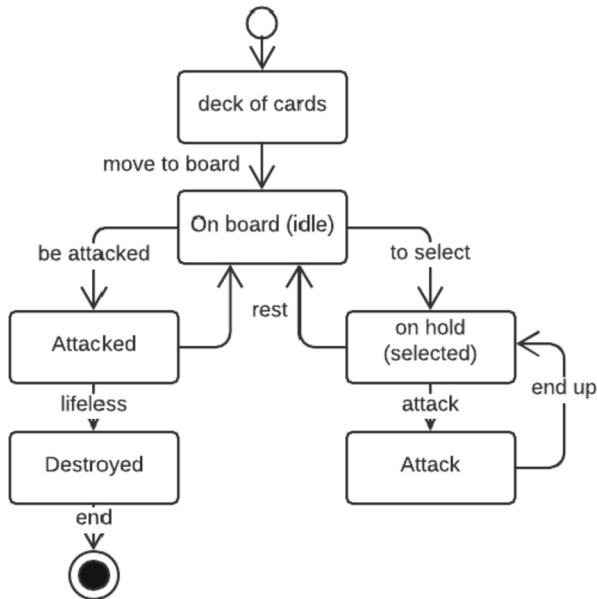


Fig. 8. The state machine for the cards.

The implementation of three levels was carried out, which correspond to the natural regions of the coast, mountains, and jungle belonging to the Tahuantinsuyo, the empire of the Incas. In each level, the difficulty increases because the available cards decrease, making it difficult for the player to restrict the possibility of drawing another card from the deck to save themselves from an enemy attack. For the first level (coast), the number of available cards is 6, for the second level (mountain), the number of available cards is 5, and for the third level (jungle), the number of available cards is 3. To pass a level, it is necessary to beat all the enemy cards on the board. The three levels are shown in Figs. 9, 10, and 11.



Fig. 9. Videogame level 1 (coast).

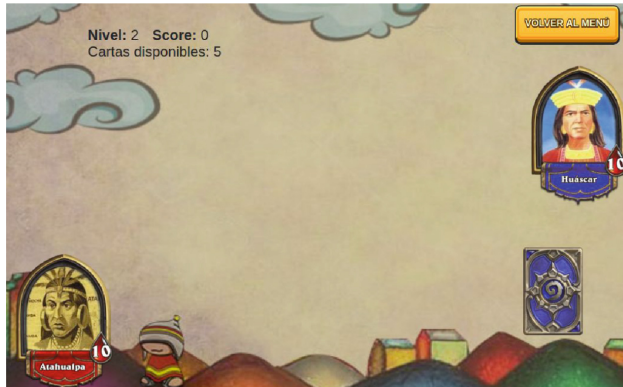


Fig. 10. Videogame level 2 (mountain).

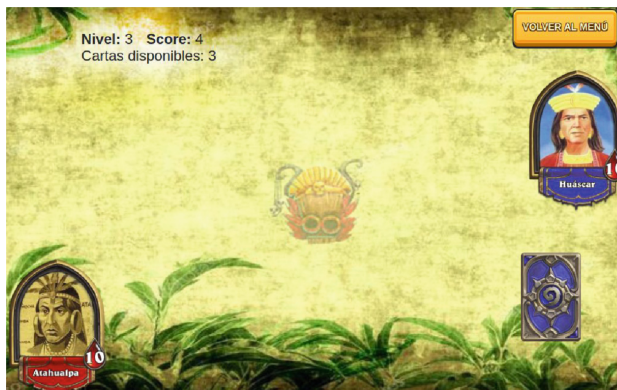


Fig. 11. Videogame level 3 (jungle).

The video game instructions were made in the main menu, where the controls are indicated, the game rules for each type of card (water, ice, and fire), the battle rules, and finally, the instructions on how to pass the level (see Fig. 12).



Fig. 12. Videogame instructions.

4 Results

For this work, three tests were carried out:

- Unit tests,
- Test of performance,
- Acceptance Tests.

4.1 Unit Tests

The tests were developed using the Jasmine tool, a JavaScript-based library for unit testing. It consists of several parts: The Corridor, Test framework, and Plugins. All unit tests were successful, the methods used within the game were correct.

In Fig. 13, the results of the unit tests are shown.

```
3 specs, 0 failures, randomized with seed 42992 finished in 0.124s

Metodo para cambiarEstadoNivel
  • Prueba de los estados de nivel del juego

Metodo para cambiarEstadoGlobal
  • Prueba de los estados del juego

Metodo para cambiarEstado
  • Prueba de cambio de estados
```

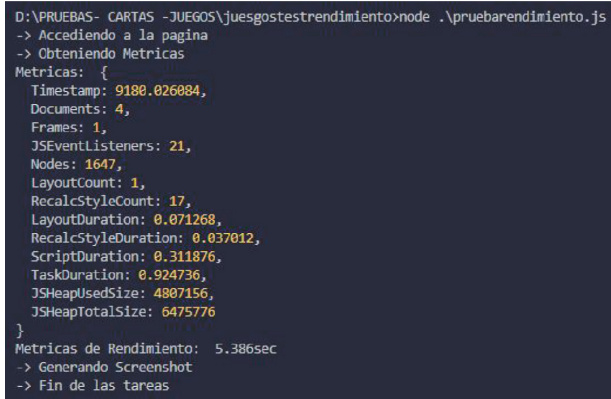
Fig. 13. Unit test results.

4.2 Test of Performance

The Performance Tests carried out were aimed at: Testing the performance of the BattleCards game and Identifying the delay time in the execution of the game. The tests were developed using the Puppeteer tool, a Node.js library. It provides a high-level API

that allows automating actions on Google browsers: Chrome and its open-source version Chromium. The results obtained for the game's loading time are 5,386 s.

In Fig. 14, the results of the Test of performance are shown.



```
D:\PRUEBAS- CARTAS -JUEGOS\juesgostestrendimiento>node .\pruebarendimiento.js
-> Accediendo a la pagina
-> Obteniendo Metricas
Metricas: {
  Timestamp: 9180.026884,
  Documents: 4,
  Frames: 1,
  JSEventListeners: 21,
  Nodes: 1647,
  Layoutcount: 1,
  RecalcStyleCount: 17,
  LayoutDuration: 0.071268,
  RecalcStyleDuration: 0.037012,
  ScriptDuration: 0.311876,
  TaskDuration: 0.924736,
  JSHeapUsedSize: 4807156,
  JSHeapTotalSize: 6475776
}
Metricas de Rendimiento: 5.386sec
-> Generando Screenshot
-> Fin de las tareas
```

Fig. 14. Test of performance results.

4.3 Acceptance Tests

The acceptance test is the last test action before deploying the software. The objective of the acceptance test is to check if the game is ready and can be used by users to perform the functions and tasks for which it was designed. For this Test, the nine user stories were carried out.

To verify the initial results, it has been based on a series of evaluation metrics to measure the quality of the game [19], test all the functions and mechanics of the game, and thus control the quality. For this, a table of the metrics to be used has been made and is detailed in Table 2.

According to what was established, multiple interviews were conducted with schoolchildren to evaluate the video game, and pre-test and post-test surveys were also conducted for the game's usability.

According to what was established, multiple interviews were conducted with schoolchildren between the ages of 10 and 12 at the primary level to evaluate the video game; these surveys were of satisfaction, pre-test and post-test; the latter to assess the game's usability.

The analysis of the satisfaction survey carried out on schoolchildren had the following results: 80% found it attractive, 60% clearly understood the dynamics of the game; In addition, they were asked about the features to improve in the game, the main features being graphic design and visual effects. Finally, 100% of schoolchildren commented that they would play it again, thus concluding that the proposed educational game is exciting.

Table 2. Result of gameplay metrics

	Nombre de la métrica	Propósito	Results
Effectiveness	Target effectiveness	What percentage of goals and challenges have been successfully achieved?	X = 1–16.66% X = 83.33%
	Goal completeness	What percentage of goals and challenges have been completed?	X = 5/6 X = 0.83
Efficiency	Finish time	How long does it take for the player to achieve a goal?	X = 7 (min.)
	Efficiency relative to the user level	How efficient is an expert player versus a new player?	X = 6/10 X = 0.6
Flexibility	Accessibility	What percentage of goals are achieved using different forms of interaction other than those used by default?	X = 0/6 X = 0

Finally, an average of all the factors described in the metrics were obtained, obtaining an initial result of 71.8% of effectiveness, of efficiency 60%, in flexibility metrics, there are no goals with different interactions, in metrics of 100% safety, 90% fault tolerance and finally 80% satisfaction.

5 Conclusions

A Battle Card game was developed based on the Inca empire and implemented with artificial intelligence algorithms, such as the seek and flee and arrive movement algorithms, the Depth First Search algorithm, and a state machine.

The Scrum framework was applied to manage the project and maintain constant communication with the development team. In addition, Phaser.io was used as a framework for game development on desktop and mobile platforms.

Tests were carried out, and the results were evaluated through game quality metrics, obtaining a result of 71.8% of effectiveness, of efficiency 60%, in flexibility metrics, there are no goals with different interactions, in metrics of 100% safety, 90% fault tolerance and finally 80% satisfaction.

The results showed that educators and specialists could use the video game “Battle Cards” as a pedagogical tool in student learning.

As future work, it is suggested to apply other artificial intelligence algorithms that optimize search and selection operations.

During the project’s development, different points to consider as future work have been identified: The design of the cards could be improved in appearance and adapted to any device since it could be developed for mobile devices for both Android and iOS.

The structure of the instructions like those of the board will have future versions depending on the difficulty levels that can be added to the game.

A virtual assistant could be added to help in the detailed explanation of the game, so users can clearly understand how the game works.

In new versions, more rivals could be added to the game.

Develop a new functionality to play online with other opponents who live in Peru or other countries in the world.

Add the multiplayer option, that is, two players against the machine.

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